

1.0. Purpose of Lab Exercise

Firearm and toolmark examiners analyze cartridge cases, bullets, and toolmarks to render a conclusion as to whether a questioned or crime scene source originated from a specific known firearm or tool. The primary instrument for this examination is the **comparison microscope**, a dual-field optical instrument that allows two specimens to be viewed simultaneously so that surface features can be aligned and evaluated side by side.

Examiners rely on their evaluation of class and individual characteristics to reach their conclusions. **Class characteristics** are features shared by all items produced by a given manufacturer to a given specification: caliber, rifling twist direction and rate, land and groove width and depth, firing pin shape, etc. **Individual characteristics** are microscopic surface features unique to a single firearm, produced at random during machining, tool wear, corrosion, and use. When individual characteristics on a questioned specimen correspond with those on a known exemplar at a level not observable in non-matching samples, the examiner may conclude that both were produced by the same source.

This laboratory exercise uses a Virtual Comparison Microscope (VCM), a browser-based simulation of the comparison microscopy workflow, to develop your ability to distinguish class from individual characteristics, systematically compare two specimens, and express a conclusion using the standard framework of the Association of Firearms and Toolmark Examiners (AFTE).

1.1. Learning Objectives

Learn the fundamentals and technique of performing cartridge case comparisons with comparison microscopy using known matches. You will also begin to understand the variability present in how a firearm marks evidence. By the end of this lab, students will be able to:

- 🗨️ **Describe** the fundamental principles of forensic firearm and toolmark examination.
- 🔍 **Examine** class and individual characteristics present in evidence.
- 🖐️ **Identify** features and determine their value in forming a conclusion.
- 🧑🔬 **Conduct** systematic side-by-side comparisons between known and questioned samples.
- 🧠 **Explain** the importance of consistency, accuracy, and documentation in forensic firearm and toolmark comparisons.
- ⚖️ **Evaluate** the limitations of firearm and toolmark evidence, including subclass characteristics and subjectivity in interpretation.

2.0. Background for the Lab Exercise

In 1985, the Criteria for Identification Committee formalized the AFTE Theory of Identification as it Relates to Toolmarks. The theory articulates three principles that provide the conceptual basis for comparing toolmarks for the purpose of identifying them as having a common source:

1. The theory of identification as it pertains to toolmarks enables opinions of common origin to be made when the unique surface contours of two toolmarks are in **sufficient agreement**.

2. This sufficient agreement is related to the significant duplication of random toolmarks as evidenced by the correspondence of a pattern or combination of patterns of surface contours. Significance is determined by the comparative examination of two or more sets of surface contour patterns comprised of individual peaks, ridges, and furrows. Specifically, the relative height or depth, width, curvature and spatial relationship of the individual peaks, ridges and furrows within one set of surface contours are defined and compared to the corresponding features in the second set of contours. Agreement is significant when it exceeds the best agreement demonstrated between two toolmarks known to have been produced by different tools and is consistent with agreement demonstrated by toolmarks known to have been produced by the same tool. The statement that sufficient agreement exists between two toolmarks means that the likelihood another tool could have made the mark is so remote as to be considered a practical impossibility.
3. The current interpretation of individualization/identification is subjective in nature, founded on scientific principles and based on the examiners training and experience.

Ultimately, sufficient agreement is the product of the examiners personal training, skills, and experience in:

- recognizing corresponding patterns of matching striations,
- recognizing corresponding patterns within impressed toolmarks,
- determining the best known non-match in their personal experience,
- training and experience in comparing striated and impressed toolmarks.

2.1. AFTE Range of Conclusions

Based on the AFTE Theory of Identification, there are four categories of examination outcomes (Range of Conclusions Possible When Comparing Toolmarks) typically used by firearm examiners in the microscopic comparison of fired samples:

- Identification
- Inconclusive
- Elimination
- Unsuitable for comparison

To reach these conclusions, the most widely *accepted method* used in conducting a toolmark examination is a **side-by-side**, microscopic **comparison** of the markings on a questioned material item to known source marks imparted by a tool.

3.0. Procedure

3.1. Accessing the Virtual Comparison Microscope

1. Open a browser (Firefox or Safari recommended, *currently not working with Chrome*).
Navigate to: <https://jsspaulding.github.io/VCM-app>
2. The interface will display two image panels side by side. The left panel is the Questioned (Q) specimen field, and the right panel is the Known (K) specimen field.
3. Select the appropriate "Specimen Type" dropdown in the toolbar to begin the exercise of choice.

You will be completing the "Cartridge Case – Basic" and "Fired Bullet – Basic" comparisons.

3.2. Understanding the Interface

Locate and identify each of the following controls before proceeding.

- *Specimen Type selector (toolbar, top)*: switches the image library between Basic, Intermediate, and Advanced tiers
- *Load into selector and file input buttons*: used to load images into the Left (Q) or Right (K) field
- *Sample Gallery (side panel)*: thumbnail grid of available specimens; click any thumbnail to load it into the selected field
- *Objective selector (above each field)*: changes optical magnification (1x, 2x, 4x, 10x, 20x, 40x, 63x)
- *Stage readout (header bar of each field)*: displays current X/Y pan position in pixels and rotation angle in degrees
- *Mode selector (toolbar)*: choose Independent, Locked (both fields move together), or Overlay
- *Illumination controls (side panel)*: light angle dial, intensity, coaxial mix, and focus depth sliders. Note several samples are already illuminated and it may be easier to turn this down to a low level.
- *Image Controls (side panel)*: brightness, contrast, edge enhancement, color channel selection
- *Measurement ruler (side panel)*: click-drag in a field to measure distances between features
- *Annotation tools (side panel)*: place point markers, arrows, region-of-interest boxes, and text labels
- *Keyboard shortcuts*: A/D = rotate left/right; Arrow keys = pan; Tab = switch active field

A screen capture of the VCM user interface with labels is provided below.

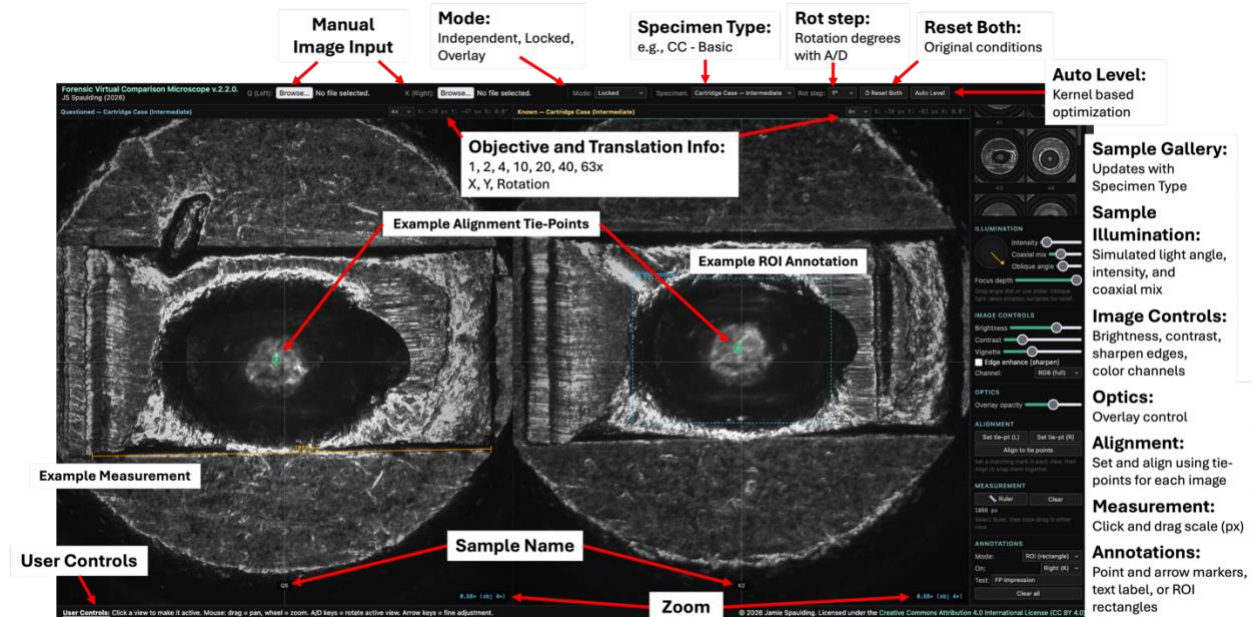


Figure 1. Labelled diagram of the VCM user interface.

3.3. Examination Procedure

Follow this procedure for pairwise comparisons in the "Cartridge Case – Basic" and "Fired Bullet – Basic" specimen types.

3.3.1. Step 1: Load the Specimen Pair

1. In the Sample Gallery, confirm the correct tier is selected in the Specimen Type dropdown.
2. Set the "Load into" selector to "Left (Q)." Click the questioned specimen thumbnail (Q1, Q2, etc.). The image will appear in the left field.
3. Set the "Load into" selector to "Right (K)." Click the corresponding known specimen thumbnail (K1, K2, etc.).
4. Confirm the sample name badge appears beneath each image in its respective field. Alternately, you can upload an image to the respective sample view from your computer using the "Choose File" buttons at the top of the interface.

3.3.2. Step 2: Initial Survey at Low Magnification

1. Set both objective selectors to a lower power (e.g., 4x) where you can visualize the entire sample. Use this magnification for your initial assessment.
2. Set the Mode selector to "Locked." Pan and zoom will now operate on both fields simultaneously, making it easier to compare equivalent areas.
3. Examine the overall appearance of each specimen. For cartridge cases: identify the firing pin impression, the breech face impressions present, and any other markings of note on the primer surface. For fired bullets, note the general appearance of the land engraved areas.
4. Appropriately document visible class characteristics (caliber, firing pin shape, rifling number/width, tool type, etc.). Assess visible damage, deformation, or obliteration.

Record any additional observations of note at this stage. Capture overall images using a screen capture tool.

The VCM interface allows you to add annotations directly to the images which you can screen capture as well.

- "Point marker" mode: on each field, click to place a marker on noteworthy features (either support or refute comparison decision)
 - "Label" mode: enter a brief label in the Annotation Text field (e.g., "FP aperture shear") and click on the marked feature to attach the label.
 - "ROI" mode: drag a rectangle around the primary area of interest in each field and the label will be attached.
5. While under low magnification, orient the surfaces such that they are aligned. The alignment tool can be used to set tie points in each image for alignment.
- Click "Set tie-pt (L)" and click on a distinctive striation mark in the left field. Click "Set tie-pt (R)" and click on the corresponding mark in the right field.
 - Click "Align to tie points." The left field will translate so the two marks coincide. Examine the surrounding area for correspondence of adjacent features.

3.3.2. Step 3: Illuminate for Surface Relief

Important note: some samples were captured with coaxial and/or oblique lighting and may be best visualized with an intensity of zero.

1. Switch the Mode selector to "Independent."
2. In the Illumination panel, adjust the Light Angle slider or click the angle dial to position the light at approximately 45°. Observe how striations and surface impressions/texture become visible as light and shadow relief.
3. Adjust the Intensity slider (start low and increase as needed) and Coaxial Mix slider (start at 25–30%). The goal is to maximize the visibility of individual marks without losing detail in shadows.
4. Apply the same illumination settings to both fields. Note that different specimens may benefit from slightly different angles, you may adjust independently per field.

3.3.4. Step 4: Systematic Examination at Higher Magnification

1. Increase the objective power for detailed examination.
2. Work through each major feature area systematically and capture micrograph captures of significant features:
 - a. *Breechface impressions:* marks impressed into the primer and surrounding surface by machined imperfections on the firearm's bolt face. Examine the pattern, density, and angular orientation.
 - b. *Firing pin impression:* marked impression from the firing pin engaging the primer surface. Also note firing pin drag marks which are elongated tear-like scrapes extending from the firing pin's indentation on the primer. These occur when the slide begins to recoil and unlock, pulling the cartridge casing away before the firing pin has fully retracted into the breech block.
 - c. *Firing pin aperture marks:* marks around the firing pin impression left by the edges of the firing pin aperture in the breech block. Note the shape, depth, and

any asymmetries. There may be shear marks left on the primer which occur when the primer flows backward under pressure and is physically sheared or scraped against the edges of the firing pin aperture as the firearm cycles

- d. *Ejector marks*: Impression or scrape produced by the ejector as the case is expelled. Note location, shape, and any associated marks. Note: not present on all CC sample images.
3. For each feature area, rotate and pan the specimens to align the corresponding region in both fields. Use the A/D keys for fine rotation.
4. Perform Side-by-Side Comparison:
 - a. Align questioned and known items in the comparison microscope
 - b. Compare class characteristics for consistency
 - c. Evaluate individual characteristics for agreement or disagreement
 - d. Rotate, tilt, and adjust lighting to reveal striated or impressed detail
 - e. Document all observed similarities and differences. See note on Annotation tool in Step 4.
 - f. Once aligned, use of the "Locked" function mode will move the samples simultaneously which will assist in evaluation of feature prevalence on both samples.

3.3.5. Step 5: Reach and Document Conclusions

1. Based on your examination, reach a conclusion using the AFTE framework:
 - a. **Source Identification**: conclusion that two toolmarks originated from the same source. This conclusion is an examiner's opinion that all observed class characteristics are in agreement and the quality and quantity of corresponding individual characteristics is such that the examiner would not expect to find that same combination of individual characteristics repeated in another source and has found insufficient disagreement of individual characteristics to conclude they originated from different sources.
 - b. **Inconclusive**: conclusion that all observed class characteristics are in agreement but there is insufficient quality and/or quantity of corresponding individual characteristics such that the examiner is unable to identify or exclude the two toolmarks as having originated from the same source.
 - c. **Source Elimination**: conclusion that two toolmarks did not originate from the same source. The basis for a 'source exclusion' conclusion is an examiner's opinion that the observed class and/or individual characteristics provide extremely strong support for the proposition that the two toolmarks came from different sources and extremely weak or no support for the proposition that the two toolmarks came from the same source.
2. Record your conclusion and a written justification citing specific observed features. Record the basis for conclusion referencing observed characteristics, include annotated captures supporting findings. Ensure your notes are complete, contemporaneous, and reproducible.
 - a. If you feel uncertain, record your observations and reasoning honestly. In professional practice, an Inconclusive conclusion is appropriate and important, it

is not a failure. The quality of your reasoning is more important than reaching a definitive conclusion.

3. Proceed to the next specimen pair. Compare all samples in the "Cartridge Case – Basic" and "Fired Bullet – Basic" specimen types in a pairwise fashion. Record your answers in the attached worksheet (Section XX.0).
 - a. Pairwise comparison is the process of comparing a set of options using head-to-head pairs.

4.0. Post Laboratory Questions

1. Based on your analysis, how confident are you in the match decisions you made? What supports or undermines that confidence?
2. If called to testify as an expert, what key points would you emphasize in your explanation of the firearm evidence?
3. Evaluate the criteria set forth by the AFTE Theory of Identification. How reliable is the assessment of sufficient agreement? What sources of error are possible when drawing conclusions under these criteria?
4. In practice, examiners may occasionally reach incorrect conclusions, either a false identification (two samples share a common source when they do not) or a false elimination (negative outcome but truly from a common source). Based on your experience with the VCM exercise, describe two specific factors that could lead a novice examiner toward a false identification. How does the discipline's reliance on the examiner's judgment, rather than a quantitative threshold, contribute to this risk, and what procedural safeguards exist to mitigate it?
5. The VCM simulates several features of a physical forensic comparison microscope, but does not reproduce all aspects of the examination experience. Identify and explain two specific ways in which this virtual tool differs from a physical comparison microscope and discuss how each difference might affect an examiner's ability to make accurate comparison decisions.

6.0. Report Requirements

Collate the completed answer sheet and your answers to the post-lab questions into a single file and upload to the drop box on Canvas.

Name: _____

Date: _____

Part I: Cartridge Case – Basic

For this exercise, you will identify which of the known and questioned samples share a common source reaching the conclusion of source identification. For each “Q Sample” identify the corresponding “K Sample” and provide your observations supporting the conclusion as described above. Attach screen captures illustrating your observations as appropriate.

Note: For the Basic exercise, each Q sample should be identified to a K sample.

Q Sample	Conclusion	Observations/Notes
	K Sample	Justification/describe individual characteristics
Q1		
Q2		
Q3		
Q4		
Q5		
Q6		
Q7		
Q8		

Part II: Fired Bullet – Basic

For this exercise, you will identify which of the known and questioned land engraved areas (LEA) share a common source reaching the conclusion of source identification. For each “Q Sample” identify the corresponding “K Sample” and provide your observations supporting the conclusion as described above. Attach screen captures illustrating your observations as appropriate.

Notes:

1. The K and Q samples were fired from the same firearm. It is your responsibility to determine the LEAs which correspond to one another.
2. The order of the LEAs preserved through consistent rotation. This means that once you have identified the corresponding LEAs, the next pair should also correspond.
 - a. Example: Q1-L1 = K1-L1 $\therefore$ Q1-L2 = K1-L2, Q1-L3 = K1-L3...

Q Sample	Conclusion K Sample	Observations/Notes Justification/describe individual characteristics
Q1-1		
Q1-2		
Q1-3		
Q1-4		
Q1-5		
Q1-6		